

Mahdi Shabani

MACHINE LEARNING · SPATIAL DATA SCIENCE · MINERAL EXPLORATION

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Machine Learning researcher focused on spatial data science, explainable AI, uncertainty quantification, and computer vision for mineral exploration.

Career focus: AI for Mining and Geoscience

Data-Driven · Explainable · Uncertainty-Aware · Spatially Informed · Decision-Oriented

SKILLS & EXPERTISE

Machine Learning

Python, Machine Learning, Deep Learning, Scikit-learn, PyTorch, Feature Engineering, Model Evaluation, Hyperparameter Optimization

Spatial Data Science

Spatial Data Analysis, GIS Analysis, Geospatial Machine Learning, Spatial Modeling, Mineral Prospectivity Mapping

AI Research

Explainable AI (XAI), Uncertainty Quantification, Uncertainty-Aware Machine Learning, Interpretable Machine Learning, Decision Support Systems

Computer Vision

Computer Vision, Image Classification, Transfer Learning, Active Learning, Drill Core Image Analysis

Data Science

Exploratory Data Analysis, Statistical Analysis, Data Visualization, Geochemical Data Analysis

EXPERIENCE & EDUCATION

B.Sc. in Chemistry

University · 2020 — 2024

M.Sc. in Industrial Engineering

Khwarizmi University — Modeling and Data Analysis · 2024 — 2026

Research & Projects

Independent Research — AI · Spatial Data Science · Mineral Exploration · 2025 — Present

PROJECTS

AIM Core Log

AI-Powered Drill Core Classification

In progress · Mineral Exploration · Computer Vision

An AI-powered computer vision system designed to automate lithology classification from drill core photographs for mineral exploration. The system uses a PyTorch-based training pipeline with transfer learning and an active-learning feedback loop that incorporates geologist corrections to continuously improve model performance.

Tech: Python, PyTorch, Computer Vision, Transfer Learning, Active Learning, Deep Learning

Impact: Reducing manual core-logging workload while creating a scalable and continuously improving geological classification workflow.

KanYab

AI-Powered Geochemical Exploration Analytics

In progress · Mineral Exploration · Geochemical Data Science

An AI-powered analytics system for mineral exploration that identifies anomalous geochemical samples, discovers meaningful element relationships, and highlights pathfinder signatures associated with mineralization.

Tech: Python, Machine Learning, Geochemical Data Analysis, Anomaly Detection, Statistical Analysis, Data Visualization

Impact: Helping exploration teams transform large geochemical datasets into interpretable evidence for prioritizing exploration targets.

RESEARCH

Development of an Explainable and Uncertainty-Aware Machine Learning Framework for Spatial Decision-Making

Research · Explainable AI · Uncertainty Quantification · Spatial Decision-Making

A research framework that integrates explainable artificial intelligence and uncertainty quantification into spatial machine learning workflows. The goal is to improve transparency, reliability, and decision confidence when machine learning models are applied to complex spatial problems.

Areas: Explainable AI, Uncertainty Quantification, Spatial Machine Learning, Spatial Decision-Making, Decision Support Systems

RESEARCH INTERESTS

Explainable Artificial Intelligence, Uncertainty-Aware Machine Learning, Spatial Machine Learning, Geospatial Data Science, Mineral Prospectivity Mapping, Intelligent Mineral Exploration, Computer Vision for Geoscience, AI-Assisted Geological Interpretation, Spatial Decision Support Systems